01 Design explanation for the project team

Fish skin to material-grade collagen

Model rationale, parameter provenance and industrial scale-up
29 October 2026

The design compares two purification alternatives after a common lumpfish-skin pretreatment, acid extraction and salting-out section. **Dialysis_0.4** is the reference route. **Washing_0.4** replaces redissolution/dialysis with eight reslurry-and-filtration cycles. Both deliver a specified powder endpoint.

The project has three connected calculation layers: Aspen carries composition and conditional sensible-energy accounting; external transport calculations estimate washing, filtration and drying performance; the industrial design allocates physical equipment, operating time and utilities. These layers answer different engineering questions.

Level	Basis	What to explain
Native Aspen models	270.27027 kg wet skin/batch;10 kg collagen at purification inlet	14 Washing blocks;17 Dialysis blocks. Exact active inputs are retained.
Physical transport models	Same native inventory	Finite-region wash exchange, pressure filtration, freezing, sublimation, desorption and cold-trap loads.
Industrial design V021	18 native batches/physical lot;200 t/y contained-collagen target	Washing equipment,1,145-lot campaign, piping, thermal services, pump candidates, utilities and layout allocation.

Use the companion parameter register to locate an individual input. It contains 2852 parameter/specification records, including source values, calculated quantities and TBD fields. Repeated route/specification records are not independent material properties. All originals are preserved. This explanation does not change or rerun Aspen.

A useful opening statement

“We built a traceable conditional process design. The extraction recipe and composition have a lumpfish literature basis. We compared purification alternatives, then linked the material balance to equipment and an industrial campaign. We distinguish direct source measurements, transferred analogues and design assumptions.”

02 How to trace a number and explain a design decision

Classification	Meaning	Example
Literature datum	A reported value for a defined experiment/material	Scheja Table 2 wet/dry/collagen masses.
Transferred analogue	A published value used for a different material or duty	UCL compressed-collagen permeability applied as a fish-cake stress proxy.
Design assumption	A selected operating, geometry or constitutive input	85% held liquor,3 mm diffusion length,0.1%/30 min collagen release,-35 C frozen critical temperature.
Calculated result	Follows from named inputs and equations	Alpha, per-component Aspen split, dryer cycle, pump head or annual output.
Manufacturer information	Published hardware geometry or performance at stated conditions	LKH water 20 C pump curves; DOWFROST properties.
TBD	Required input has not been established	Dialysis membrane area, actual cake test, final nozzles, vendor minimum flow.

Trace in this order

Find the parameter and its route/revision. Read the exact source locator. Check whether the number is reported, transferred, selected or calculated. Follow the equation to its inputs. State the resulting equipment requirement and the conditions under which it applies.

What “every parameter” covers

The register includes 268 native Aspen setting records, grouped unused NC coefficient zeros, source-level recipe assumptions, active transport settings, refrigeration inputs, native-batch equipment specification fields and V021 industrial parameters. It carries all 546 existing hydraulic/thermal source rows. It identifies historical UF/drying assumptions separately. Proprietary databank internal coefficient tables and every historical sensitivity output are not reprinted.

Reference IDs L01-L11 identify papers or scientific data. T01-T06 identify textbook/course methods. P01-P03 identify property references. V01-V06 identify manufacturer methods. A01/A02 identify Aspen evidence. D01 and D21 identify project choices. Codes inherited from other workbooks carry their own V014/V021/HISTORY namespace in the source dictionary.

03 The two routes and why they exist

Section	Both routes	Where they differ
Pretreatment	Skin + alkali, drain, six water rinses	Same feed, imposed net retention and reagent basis.
Extraction	Prepare 0.9 M HAc, mix with cleaned skin, condition 21 C, hold 36 h, clarify	Same empirical extract collection.
Salt-out	Add nominal 2.6 M NaCl, cool 4 C, hold, initial separation	Same SALTCAKE interface.
Purification	Remove salt/acid with limited collagen loss	Dialysis: redissolve then exchange with baths. Washing: repeated reslurry/filter operations.
Drying	Freeze-dried powder mass endpoint	Very different water loads and therefore different external dryer requirements.

L02 explicitly investigates replacing conventional dialysis with washing for yellowfin-tuna collagen. This is the source of the process alternative. It is not evidence that both routes must run simultaneously or that a plant needs two complete purification lines. Lumpfish extraction conditions are transferred from L01. The combined project process is an engineered synthesis of sources.

The route comparison fixes raw-skin inventory and the 10 kg purification-inlet collagen basis. It does not fix recovered collagen exactly. Comparisons per kg product must use each route’s own recovered mass. Product functionality is not established merely by matching model composition.

Other explored routes

Early work also screened UF/DF, wet delivery, drying alternatives and condensate reuse. They are historical or optional studies, not hidden blocks in the two current 0.4 files. Reuse savings must not be added to the current once-through water balance unless the recycle is explicitly integrated and rerun.

04 Component definitions and physical properties

Component	Representation	Reason and consequence
WATER; HAC	Conventional MIXED components; NRTL-HOC; VLMX26 volume rule	Molecular water/acetic-acid nonideality and acid vapour association. Some binary parameters are extrapolated at 4 C below their recorded 20 C lower limit.
COLLAGEN	NC, ENTHGEN and DNSTYGEN	Tracks nonvolatile collagen mass and approximate sensible heat. Does not predict dissolution, conformation or viscosity.
OTHERDRY	NC with the same generic protein property analogue	Unidentified native dry material, including possible ash/fat. It is not a measured noncollagen-protein fraction.
BASEEQ; NACLEQ	NC apparent equivalents	Retain nonvolatile reagent equivalents. Solid-reference Cp/density are bookkeeping analogues, not solution partial properties.

Cp_protein(tC) = 2008.2 + 1.2089 tC - 0.0013129 tC^2 [J/(kg K)]

rho_protein(tC) = 1329.9 - 0.51840 tC [kg/m3]

T01 supplies generic food-protein correlations. Aspen inputs use Kelvin: HCGEN first four coefficients are [1580.0322848498,1.926137270,-0.00131290,0]; DENGEN coefficients are [1471.500960,-0.51840,0,0]. The remaining 76 entries are zeros for unused generic constituents. This conversion prevents a273.15-degree temperature-origin error.

BASEEQ Cp=1488.41847 J/(kgK), density 2130 kg/m3. NACLEQ Cp=864.111134 J/(kgK), density 2160 kg/m3. P01/P02 give the solid analogues and their reference temperatures. They do not supply neutralization or salt-dissolution heat.

No electrolyte pH, salt-out equilibrium, membrane rejection, collagen denaturation or film properties are predicted by changing the property-method name. The parameter choices are sufficient for the declared accounting model; their limits matter for physical interpretation.

05 Apparent equivalents and the absence of a kinetic reactor

The external composition ledger uses seven quantities: collagen, other dry matter, water, free HAc, NaOH, anhydrous sodium acetate and NaCl. Aspen uses six coordinates. Sodium acetate is represented by conserved acid/base/water equivalents rather than an explicit electrolyte component.

n = m_NaAc / MW_NaAc

BASEEQ = m_NaOH + n MW_NaOH

HAC_app = m_HAc + n MW_HAc; WATER_app = m_water - n MW_water

Molecular weights in the compatible g/mol registry are WATER18.0153,NaOH39.997,HAc60.052,NaAc82.0337 andNaCl58.443. Their stoichiometric closure preserves total mass. The apparent coordinates do not reproduce real electrolyte density, enthalpy or species distributions.

What this means for reactions

No RStoic, RCSTR or RPlug is present in either current main-process model. Acid extraction is represented by mixing, conditioning and empirical collection. Chemical neutralization can occur physically, but the equivalent bookkeeping does not independently calculate reaction equilibrium, rate or heat. The project also does not model collagen hydrolysis into peptides.

An agitated batch extractor is chosen because the process contacts fibrous skin with acid for a specified time and temperature, then separates residual solids. Calling its physical vessel a “reactor” does not make EXTRACT a kinetic reactor. The actual Aspen EXTRACT block is HEATER.

Important interpretation

Apparent HAC includes acetate-bound acid equivalent. Raw NC fraction is not the true nonvolatile dry fraction. Product purity/moisture and free-acid removal must use the controlled decoded/external ledger. Inversion is conditional on the chosen neutralization convention; simultaneous free acid/base traces cannot be arbitrarily reconstructed from six coordinates.

06 Raw-skin basis and normalization: a worked source chain

L01 Table 2 provides a compatible 34 kg composition branch: wet skin 34000 g,dry matter 4862 g,collagen 3118 g. Those data define the three native feed lumps. They are not mixed with a separate proximate-analysis specimen or forced to close inconsistent pilot table rows.

Dry fraction = 4862/34000 = 0.143

Collagen fraction of native dry matter = 3118/4862 = 0.641299877

Quantity	Native batch result	Origin
Wet raw skin	270.27027027 kg	Scale the upstream model so collected extract contains 10 kg collagen.
Water	231.62162162 kg	Wet mass x(1-0.143).
Collagen	24.785373609 kg	Wet mass x0.143 x3118/4862.
OTHERDRY	13.863275040 kg	Dry matter minus collagen.
Collected extract collagen	10 kg	Chosen purification comparison normalization; not a measured factory batch.

After pretreatment/rinsing, modeled collagen is 23.13990461 kg. Empirical extraction/collection fraction 1258/2911=0.432153899 gives 10 kg in EXTRLIQ. The raw-skin batch is therefore determined by this selected yield chain. Ten kilograms is not the raw collagen content and 97.09% purification recovery is not 97.09% recovery from the raw fish skin.

The two 0.4 files use the same upstream basis. The Aspen label kg/h represents one batch over one numerical hour. A7211 W native duty corresponds to 7.211 kWh/batch under that convention; it is not a7211 W continuous factory heating requirement.

07 Alkaline contact and skin drainage

PRETREAT - MIXER

Native connections: SKIN CAUSTIC to PREPSLUR.

Physical interpretation: T-101 agitated wash/contact tank. Mix raw skin and prepared alkali; six-hour contact occurs in the batch schedule.

Why this block: A mixer accounts for mixing. It does not model contaminant hydrolysis or collagen reaction kinetics.

DRAIN - SEP

Native connections: PREPSLUR to ALKCAKE SPENTALK.

Physical interpretation: S-101 screen/drain function, potentially integrated in T-101. Separate retained skin and held liquid from SPENTALK.

Why this block: SEP imposes component allocation; a real screen/basket/drain requires particle and drainage data.

Parameter	Value	Basis
NaOH concentration / liquor ratio	0.1 mol/L;5 L/kg raw skin	L01 Sections 3.4-3.5.
Contact temperature / hold	4 C;6 h	L01 method; time implemented in external schedule.
Prepared alkali inventory	1351.35135 kg/batch	Selected 1 kg/L solution-density approximation.
NaOH-equivalent charge	5.405 kg/batch	Concentration x volume x rounded MW.
Net collagen retention after alkali and washes	2911/3118=0.933611289	L01 Table 2.
Net OTHERDRY retention	(4200-2911)/(4862-3118)=0.739105505	Derived from the same Table 2 branch.

DRAIN alone uses collagen split 0.948931349, rather than the net 0.933611289. The difference allows subsequent modeled rinse losses while reproducing the selected net pretreatment result. This is a calibrated net mass allocation, not a fit to time-dependent leaching measurements. [L01,D01]

08 Six skin rinses and acid preparation

SKWASH6 - SEP

Native connections: ALKCAKE SKWATER to CLNSKIN SKRINSE.

Physical interpretation: Repeated operation in T-101 with drainage. Combine six wash/draw operations into one equivalent separation.

Why this block: One aggregate calculation block avoids six duplicated computational stages; it is not a one-pass separator.

ACIDMIX - MIXER

Native connections: ACIDWAT HACMAKE to ACIDHOT.

Physical interpretation: T-102 acid preparation tank. Mix ACIDWAT and HACMAKE.

Why this block: Physical mixing, not a kinetic reaction vessel.

ACIDCOOL - HEATER

Native connections: ACIDHOT to ACID09 M.

Physical interpretation: Temperature-control interface on T-102. Specify ACID09 M at 21 C.

Why this block: Almost zero native duty; the name does not require a separate cooler.

Six skin rinses use 5 kg water/kg raw skin each, giving 8108.108108 kg/batch in SKWATER. The source says to wash to a pH endpoint but does not provide this water quantity or count. The model selects the smallest integer count satisfying residual free-base moles/fresh extraction-acid moles <=1e-5 under its mixing/hold-up assumptions. It does not calculate or prove pH7. [L01,D01]

A water-hold proxy(50000-4200)/4200=10.9047619 kg water/kg net dry skin comes from L01 Table 2. Fully mixed rinse/drain balances and gross-leaching allocation produce the equivalent SKWASH6 splits. Six serial operations can use one tank/drain arrangement.

The acid recipe is 0.9 M at 2 L/kg modeled washed wet skin, using a chosen 1 kg/L prepared-solution density. The charge is 751.950295 kg water plus 42.96249245 kg acid. ACID09 M is 21 C despite the historical ACIDCOOL name. Its near-zero conditioning duty does not justify a separate standalone cooler automatically. [L01 Sections 3.3-3.4]

09 The actual extraction model and vessel choice

EXTMIX - MIXER

Native connections: CLNSKIN ACID09 M to EXTSLUR.

Physical interpretation: T-201 extraction vessel. Combine CLNSKIN and ACID09 M.

Why this block: Can share the physical vessel with EXTRACT.

EXTRACT - HEATER

Native connections: EXTSLUR to EXTRDONE.

Physical interpretation: T-201 jacketed/agitated batch extractor. Compute conditioning duty; external schedule holds 36 h.

Why this block: HEATER is a thermal surrogate. No RCSTR/RPlug reaction rate, solubility model or denaturation model is present.

CLARIFY - SEP

Native connections: EXTRDONE to EXTRLIQ RESIDUE.

Physical interpretation: S-201 clarification filter/centrifuge function. Send empirically collected collagen to EXTRLIQ and remainder to RESIDUE.

Why this block: Net extraction plus collection yield is encoded here; SEP does not identify centrifuge speed or filter area.

The source recipe uses 21 C and 36 h in 0.9 M acid. A jacketed agitated batch vessel provides temperature control and contact with fibrous skin. EXTMIX and EXTRACT are two calculations that can share T-201. There is no reason to select a distillation column, equilibrium flash or a residence-time kinetic reactor for this represented operation.

Collected collagen fraction = 1258/2911 = 0.432153899

Collected OTHERDRY fraction = (1493-1258)/(4200-2911) = 0.182311870

These are net collected ratios from L01 Table 2, transferred to the selected warm recipe. They are not an independently validated 21 C single-step kinetic yield. Retained residue is calculated by conservation; inconsistent source residue/extract rows are not forced simultaneously.

CLARIFY water/HAc fraction 0.961437470 follows the assumed 2 kg water/kg retained native dry residue. Its collagen split combines extraction and collection in one endpoint. A vendor filter or centrifuge must meet that collection requirement; Aspen SEP alone cannot distinguish the best technology.

L03 reports 16.1 C solution rheological denaturation under its own test conditions. This flags a native-structure question for 21 C extraction. It does not prove our formulation denatures, nor justify changing temperature while retaining the same empirical yield.

10 Salt addition, cooling and initial separation

SALTMIX - MIXER

Native connections: EXTRLIQ SALT to SALTED.

Physical interpretation: T-301 salting vessel. Add the nominal 2.6 M salt charge.

Why this block: Salt-out equilibrium is not calculated by this mixer.

CHILL - HEATER

Native connections: SALTED to SALTCOLD.

Physical interpretation: J-301 jacket/coil on T-301. Provide sensible cooling to the selected salting state.

Why this block: Can share T-301 with SALTMIX; precipitation hold is external.

SALTOUT - SEP

Native connections: SALTCOLD to SALTCAKE BRINE.

Physical interpretation: S-301 initial separator. Retain precipitated material and mother liquor as SALTCAKE.

Why this block: SEP is chosen because capture/entrainment are supplied empirically; no thermodynamic crystallization or settling prediction.

The nominal 2.6 M salt addition comes from L01 laboratory method. A selected carrier density 1.1 kg/L closes the final-volume salt-charge calculation, giving 178.596723 kg NaCl-equivalent/batch. Salt quantity is not obtained from a salt-out equilibrium curve. The 24 h hold used in scheduling is a transferred tilapia-process choice from L11, not a measured lumpfish precipitation time.

Wet-cake liquid fraction h=0.85; held mother liquor/dry pellet = h/(1-h)=5.66667

Overall C split = 0.99 + (1-0.99)*0.04414921845 = 0.99044149218

Overall OTHERDRY split = 0.10 +0.90*0.04414921845 = 0.13973429661

The 99% collagen pellet capture and 10% OTHERDRY pellet capture are project assumptions. Mother liquor carries additional dissolved material into the wet cake. Therefore the Aspen collagen split is slightly greater than 0.99. Salt/water/acid follow the common held-liquor fraction. [L10,D01]

T-301 can combine SALTMIX and CHILL. S-301 represents initial solid/liquid separation. Sharing S-301 with the later wash filter is a possible equipment arrangement, but the initial 1304.754 kg native slurry separation is not included in the eight-round wash calculation and cannot simply be deleted.

11 Dialysis reference: redissolution and bath handling

REDMIX - MIXER

Native connections: SALTCAKE REDWATER REDACID to REDHOT.

Physical interpretation: T-401 agitated redissolution vessel. Mix cake with selected 0.5 M acid solution.

Why this block: No thermodynamic collagen dissolution or mixing-time model.

REDCHIL - HEATER

Native connections: REDHOT to REDISS.

Physical interpretation: E-401 or T-401 cooling interface. Provide sensible conditioning.

Why this block: Not a second reaction stage.

BATHMIX - MIXER

Native connections: BATHWAT BATHAC to BATHHOT.

Physical interpretation: T-402 bath preparation/storage. Aggregate acid bath and water-bath supplies.

Why this block: Bath inventory represents multiple sequential charges, not one simultaneous vessel fill.

BATHCOOL - HEATER

Native connections: BATHHOT to BATHCOLD.

Physical interpretation: E-402 bath cooler. Condition sequential bath inventory.

Why this block: Physical peak power depends on preparation schedule, not the combined Aspen inventory alone.

L02 Section 2.5.1 supports 0.5 M redissolution acid and a24 h acid-bath period followed by 24 h water, changed every 12 h. Our 100 L solution/kg cake collagen and bath/carrier ratio 5 are separate design choices. Four sequential bath exchanges are represented, first two with 0.1 M acid, then two water baths.

The model combines all fresh bath inventory into BATHWAT/BATHAC for accounting. This does not imply one tank must hold all four baths at once. Physical preparation vessels and cooler peaks need a charge-by-charge schedule. The redissolved feed has much more water than the washed cake and therefore creates a much larger dryer load.

12 Dialysis reference: what DIALYSE calculates

DIALYSE - SEP

Native connections: REDISS BATHCOLD to FDFEED DIALOUT.

Physical interpretation: M-401 dialysis membrane assembly and bath handling. Retain collagen while exchanging acid/salt with four external baths.

Why this block: SEP implements a finite-bath endpoint. Membrane area, permeability and 48 h kinetic feasibility are not independently predicted.

At an ideal finite-bath exchange: c_equilibrium = (m_feed + m_bath)/(V_feed + V_bath)

m_retained,j = c_equilibrium,j * V_feed; repeat for each fresh bath

The implementation uses dilute carrier-mass capacities as a volume approximation. With a clean five-volume bath and an unchanged feed volume, an ideal freely passing tracer would retain 1/6 per exchange. The actual ledger accounts for each acid/water bath and carrier inventory. Four exchanges remove small solutes even though the final native SEP contains only two outlets.

The imposed total collagen retention is 0.99 and OTHERDRY retention 0.50. Neither follows automatically from a membrane molecular-weight cutoff. The aggregate DIALYSE water fraction 0.04666263636 applies to the combined redissolution-plus-all-bath inlet water, not to only the collagen-side solution. It is not a membrane permeability.

Output / requirement	Status
FDFEED water	1008.07389 kg/native batch.
Final product collagen	9.80537077 kg/native batch;98.05% of 10 kg purification inlet.
48 h bath protocol	Source-method timing transferred to design.
Membrane area, permeability and mass-transfer coefficient	TBD. Finite-bath endpoint does not demonstrate 48 h equilibration.
Industrial same-output dialysis plant	Not completed in V021; reference cooler calculations are available.

A membrane assembly with bath handling is the physical device. SEP is an endpoint surrogate; a Flash block would calculate the wrong physical mechanism. A dedicated kinetic dialysis model would require membrane area, thickness/permeability, boundary-layer coefficients and verified retentate retention.

13 Washing improvement: transport and loss

WATCOOL - HEATER

Native connections: PWATER to PWAT4.

Physical interpretation: E-501 cooler and T-501 cold buffer. Prepare the total wash-water inventory.

Why this block: Actual bank area and conditioning window are external; the stream is not a continuous dose at its numerical kg/h.

CAKEWASH - SEP

Native connections: SALTCAKE PWAT4 to FDFEED PWASTE.

Physical interpretation: F-301 wash/filter bank. Use staged washing to remove acid and salt with limited collagen loss.

Why this block: External transport model supplies aggregate fractions. Industrial banks/modules are a later physical arrangement.

The current design is eight rounds,2 L water/kg then-current wet cake and 15 min contact per round. This is an engineering candidate developed from L02's repeated-washing concept; it is not the original paper's10-round,1:5,30 min protocol. PWATER totals 1064.211454 kg/native batch.

dM_s/dt = -k_release M_s; k_release = -ln(1-f_30 min)/1800

k_C = 5.55833519e-7 /s; k_OTHERDRY = 5.58351992e-6 /s

The low-loss case selects 0.1% collagen release and 1% OTHERDRY release per 30 min. Release occurs continuously during both contact and filtration. Released material enters mobile liquid; it is then retained in pore liquid or discharged. Loss is not deleted from the balance or counted twice.

Two liquid regions represent mobile and stagnant/occluded liquor. The stagnant fraction 0.10 is assumed. Species exchange continues during drainage, whose duration is predicted by the hydraulic calculation. Final wet cake, filtrate and dissolved/solid inventories are summed and compiled into CAKEWASH component fractions. [V007 coupled_wash.py]

The saved JSON case_id contains an inherited N09 label. The actual eight-element wash history and wash_rounds=8 are authoritative. This is another reason to explain the numerical settings rather than infer process design from filenames.

14 How diffusion and permeability enter washing

D_salt,4 C = 9.42211471e-10 m2/s; D_eff = hindrance * D_salt

tau = L_diff^2 / (pi^2 D_eff)

Quantity	Value	Source and interpretation
Na+/Cl- diffusion at 25 C	1.33e-9 /2.033e-9 m2/s	L05 dilute-ion database. Temperature correction gives ambipolar 4 C value.
Hindrance / diffusion length	0.20 /3 mm	Project assumptions. Aggregate/slab length differs from full cake thickness.
Salt exchange time	4839.0976 s	Calculated from the anchor and assumptions.
Acid exchange time	9678.1952 s	Assumed acid diffusion is 0.5 times salt.
Collagen/OTHERDRY exchange time	483909.7596 s	Assumed protein mobility 1/100 of salt; not a measured collagen diffusivity.
Permeability proxy k	1.18e-16 m2	L04 mean 2.82e-16 min us reported spread 1.64e-16; a stress selection.

The scientific data constrain the order of magnitude and the form of the calculation. They do not uniquely identify diffusion length, hindrance, release or retained-liquid fraction. A fit requires matched concentration/time or pressure/filtrate data. Total route recovery and a solubility-versus-salt plot are insufficient to identify all these parameters independently.

L03 informs qualitative salt-solubility and solution thermal context. It is not used to fabricate an absolute lumpfish release-rate measurement. A lower release rate and a more permeable cake improve different parts of performance and must remain distinct inputs.

15 Filtration: from resistance to area and time

epsilon = [h/(1-h)*rho_s/rho_l] / [1+h/(1-h)*rho_s/rho_l]

alpha = 1/[k rho_s (1-epsilon)] = 5.43000628e13 m/kg

t = mu alpha m_s V/(2 deltaP A^2) + mu Rm V/(deltaP A)

Input	Native value	Origin
A;deltaP	12 m2;100 kPa	Trial design selections.
mu;rho_l;rho_s	0.00157 Pa s;1000;1350 kg/m3	Water-like/solid-density proxies.
h;porosity	0.85;0.884393064	Selected liquid fraction; derived porosity.
Rm;n;void exponent	1e10/m;0.4;0.15	Unmeasured medium resistance and pressure-dependence choices.
First retained solids/filtrate	10.05470 kg /0.1347048 m3	Coupled-model outputs.

T02 supplies the Darcy/Ruth framework. Textbook resistance per thickness r has units 1/m2; mass-specific alpha has unitsm/kg. Confusing them creates a large dimensional error. The final coupled model updates solid release, growing deposited-cake resistance and pore exchange over 256 drainage intervals. [L06,T02,D01]

The simplified first-round time is 1.11416 h versus 1.11706 h in the coupled model. Eight contacts plus filtrations total 10.74308 h. Final purification recovery is 97.0876% of the 10 kg inlet. These results are conditional on the assumed release and transferred permeability.

At12 m2, first cake depth is about 5.37 mm. An equivalent one-circle area is about 3.91 m diameter. Thin-cake reslurry/unloading and achievable pressure must be a real equipment requirement. A large area alone is not evidence of a conservative or practical filter.

16 Freeze drying: physical model and primary drying

FREEZDRY - SEP

Native connections: FDFEED to PRODUCT FDCOND.

Physical interpretation: FD-401 washing / FD-411 dialysis package. Allocate water/acid to FDCOND, retain dry material in PRODUCT.

Why this block: SEP is mass accounting. External freezing/front/desorption and refrigeration calculations supply physical cycle and loads.

A freeze-dryer package provides shelves/chamber, ice condenser, refrigeration, vacuum and controls. The native SEP only implements product/condensate masses. It remains at the accounting reference state; it does not perform a real 5 Pa sublimation calculation. Physical calculations are external.

dH/dt = d/dx[k dT/dx]

Freezing uses a1 D apparent-enthalpy model,32 cells and 10 s time steps. Shelf starts at-40 C; the assumed phase interval is-2 to-1 C and the endpoint is approximately-39 C. Frozen water fraction 0.95 and thermal constants are inherited approximations, not measured formulation properties.

R_p(s) = R0 + R1*s/(1+R2*s)

J = [p_ice(T_interface)-P_chamber]/R_p

(T_shelf-T_interface)/(1/h_contact+L_frozen/k_frozen) = J L_sub

R0=5e4 Pa m2 s/kg,R1=2e8 Pa m s/kg,R2=100/m,h_contact=20 W/(m2 K),k_frozen=2.3 W/(mK). These are uncalibrated constitutive inputs. Murphy-Koop Eq7 supplies pure-ice pressure. The moving-front method is informed by L08/L09 and the archived reference benchmark; the MIT paper does not supply collagen-specific fitted coefficients.

At each front increment the shelf command is reduced as needed to keep the product bottom below the assumed critical temperature minus margin. Vapour flow is capped by selected equipment capacity. The code does not clamp temperature while preserving an inconsistent heat flux.

17 Freeze drying: temperature, secondary drying and capacity

Parameter / result	Washing	Dialysis
Chamber / trap	5 Pa /-60 C	5 Pa /-60 C
Assumed frozen critical / margin	-35 C /3 K	-35 C /3 K
Product-bottom primary limit	-38 C	-38 C
Selected active shelf area	16 m2	320 m2
Primary vapour capacity	3 kg/h	40 kg/h
Freeze / primary / secondary	1.703 /33.378 /4.837 h	1.594 /29.296 /12.573 h
Total including pumpdown/turnaround	43.918 h	47.463 h
Water in dryer feed	55.718 kg	1008.074 kg

k(T)=k15*exp[-Ea/R*(1/T-1/288.15)]

t_desorb = ln[(X0-Xeq)/(Xf-Xeq)]/k(T)

Secondary drying usesk 15=1/7200/s,Ea=40000 J/mol,Xeq=.005 kg/kg dry and 10 C shelf. The 13.1 C secondary screen is 16.1 C solution comparison minus 3 K, transferred as a design screen. It is not dry-powder thermal stability. The-35 C frozen critical temperature is a separate hypothetical input; solution denaturation cannot establish it.

Water endpoint targets 5 wt% product moisture;99% free-acid removal is separately imposed. Neither is a measured desorption/VLE prediction. The active-area selection is the smallest tested grid candidate meeting 48 h and scenario constraints, not a global optimum or vendor guarantee.

A colder assumed frozen critical temperature of-36 C extends calculated cycles beyond 48 h at the selected areas. The result is sensitive to an unmeasured property. Vendor ice inventory, peak vapour flow, active area and duct/vacuum performance must all correspond to the same required duty.

18 Refrigeration: actual blocks and coupling boundary

Historical utility studies built separate process/low/upper cycles. The current cold-trap calculation uses the verified V006 nine-block REFPROP cascade, CASCADE_2_VERIFIED. Its low-temperature refrigerant is ethane; the upper circuit uses ammonia with two compressors and an intercooler.

Actual block(s)	Physical purpose	Why used
LCOMP;LVALVE;LEVAP	Low-stage compression, throttling, evaporation	Pressure lift, isenthalpic expansion and specified evaporating/suction state.
UCOMP1;UINTER;UCOMP2	Two-stage upper ammonia compression with intercooling	Split compression ratio and cool between stages.
UCOND;UVALVE	Upper condensation and throttling	Reject heat and reduce pressure.
CASCADE HEATX	Condense ethane while evaporating ammonia	Explicit coupled heat exchanger, not two unrelated coolers.

Selected inputs include 75% isentropic and 95% mechanical efficiencies,5 K superheat and declared pressure-loss allowances. Low/upper pressure levels and 1000 kg/h normalized low-stage flow define the verified operating point; upper flow is adjusted for heat balance. CASCADE uses a trial 500 W/(m2 K) coefficient. These are design choices, not compressor-map calibration.

Selected native COP=0.80998237. At the native operating point the cold duty is 96.81789 kW and total compressor electricity 119.53086 kW. The recorded minimum segmented cascade temperature approach is about 9.61 K. Native block count does not determine adequacy: phase, pressure ratios, energy closure and exchanger approach are the relevant checks.

Primary cold-trap electricity = integrated primary cold load / COP_fixed_state

Peak primary cold-trap load is 2.40085 kW for native washing and 32.01373 kW for native dialysis. This covers a specified primary cold-trap boundary. Shelf cooling/heating, vacuum, heat leaks, frost/defrost and cleaning are separate. V021 process glycol 600 kW is also a separate utility. No fully integrated whole-plant Aspen refrigeration model is claimed.

19 Industrial scale-up is already designed

V021 develops the Washing route into an external industrial process and plant design. It does not overwrite the normalized Aspen recipe. An18 x physical lot preserves the selected native composition/yield endpoint. Every physical lot contains 4.864865 t wet skin,174.757719 kg recovered collagen and 187.074833 kg powder.

Collagen per lot = 18*9.70876217 = 174.75771906 kg

Required whole lots = ceil(200000/174.75771906) = 1145

Annual collagen = 1145*174.75771906 = 200097.5883 kg

Industrial design choice/result	Current basis	Provenance
Scale factor / annual target	18 /200 t contained collagen	D21 design selection.
Production campaign	330 days=7920 h	Chosen calendar including startup and final cleanup.
Uniform launch pitch	6.8052666 h	Calculated so 1145 whole lots finish within campaign.
First product / last product	133.775 h /7919 h	Schedule result; plant starts empty.
Final equipment release	7920 h	Schedule includes cleanup.
Powder output	214.200684 t/y	Calculated from own-route product composition.

This is more than multiplying annual mass. The dispatch reserves specific resources, resolves transfer/cleaning conflicts, and retains source holds and selected drying cycle. The checked timetable is deterministic; it is not a demonstrated availability or staffing model.

The equivalent full-scale Dialysis plant has not yet been closed. Its native model, thermal reference and 18 x cooler sizing are retained as reference calculations. Its complete same-output membrane sizing, resource dispatch and installed equipment comparison remain open.

20 Industrial equipment quantities: why more than Aspen blocks

Service	V021 quantity/capacity	Count rationale
Pretreatment	2 vessels	13.508 h occupancy vs 6.805 h pitch.
Extraction	6 vessels;36 m3 selected nominal each	39.940 h occupancy, incl 36 h hold; ceil(39.940/6.805)=6.
Salting	5 vessels;36 m3 selected nominal each	28.453 h baseline occupancy plus coordinated handling/cleanup.
Wash/filter	2 banks x14 modules;216 m2/bank	12.956 h occupancy requires 2 banks;216/14=15.429 m2 active/module.
Drying	7 groups x(136+136+16)m2	288 m2 active/physical lot;43.918 h occupancy requires 7 groups.21 cabinets total.
CIP	4 shared circuits	Feasible selected cleaning dispatch; not proved globally minimum.
Product-slurry transfer	3 shared channels	Chosen grouping and interval conflicts including cleaning; not one pump for every stream.

Aspen has one EXTRACT block, yet several vessels are needed because 36 h contact overlaps new batch launches. Conversely EXTMIX and EXTRACT can share each physical extractor. One wash calculation block becomes a bank of repeated physical modules, while eight rounds are repeated operations in each bank.

The native-batch V014 sheets contain earlier A/B arrangements and pump candidates. Those counts are not the current industrial equipment register. Use V021 quantities for the industrial case. A/B may denote staggered capacity, not a standby spare. A candidate pump is not automatically a purchased unit.

The selected wash modules and dryer cabinets are physical capacity requirements. Equal area/inventory splitting preserves the assumed layer loading and cycle, but actual thin-cake operability and dryer area-proportional vapour capability remain procurement requirements. [D21]

21 Industrial heat transfer and vessel design

Volume, area and thermal time do not scale by the same exponent. For geometrically similar vessels, volume grows with length cubed and jacket area with length squared. Scaling mass 18 x but area only 18^(2/3) would increase the conditioning time. V021 therefore adds explicit coil area while retaining the accepted 200 W/(m2 K) effective U and source thermal endpoint.

Service	Current industrial requirement	Calculation/source
J-201 extraction	66.039 m2 total:25.199 jacket+40.840 coil	Preserves 1.79208 h heat conditioning;36 m3/h hot water.
J-301 salting	72.252 m2 total:27.569 jacket+44.683 coil	Preserves 2.21711 h cooling;144 m3/h glycol.
Coil/vessel	4 parallel coil circuits;coil displaced volume included	Extractor minimum 27.963 m3;salter 30.598 m3 at 80%fill; selected 36 m3.
E-501 wash water	170.630 kW;24.429 m2;18 parallel paths	Aspen batch energy/2 h, explicit modular tubular geometry and two-side pressure loss.
E-401 reference	156.223 kW;27.483 m2;18 paths	Dialysis redissolution cooler reference.
E-402 reference	1102.258 kW;146.574 m2;108 paths	Sequential bath-preparation reference, not installed with washing.

Q_batch = change in model enthalpy; mean Qdot = Q_batch/t_conditioning

1/U_o = outer-area film + wall + fouling resistances; A = Qdot/(U_o*F*LMTD)

The current detailed exchanger basis is modular tubular construction, with heat-transfer coefficients and pressure drop calculated from the declared geometry/properties. Earlier plate-cooler U/area estimates are superseded. Plate heat exchangers remain a possible vendor alternative at matched duty; no unprovided plate geometry is assumed.

T04/T05 supply the method. Actual inputs include 30 vol% propylene glycol at approximately-1 C:rho 1037.74 kg/m3,mu 0.00747 Pa s,k0.4078 W/(mK),Cp3790.2 J/(kgK). V02 supplies these properties. U, film coefficients and area are calculated results or declared assumptions according to the individual row.

22 Piping, pumps and control logic

Q = transferred volume / allowed transfer time; v=4 Q/(pi D^2); Re=rho vD/mu

deltaP_f = f_D*(L/D)*rho*v^2/2; deltaP_K = sum(K)*rho*v^2/2

H = deltaZ + (P_receiver-P_source)/(rho g) + sum(deltaP_loss)/(rho g)

NPSHA = P_source_abs/(rho g) + z_source-z_pump - h_suction - Pvap/(rho g)

T05/T06 supply the hydraulic method. V021 selects 90 m3/h main liquid transfers by scaling the inherited 5 m3/h native transfer capacity 18 x. The route coordinates, elevations, lengths and fitting K values are design choices. Nominal bores use V03. The source absolute pressure is 100000 Pa, not silently replaced by 101325 Pa.

Twenty-three line cases include separate equipment/manifold losses. For L109 the wash header and worst parallel branch add 18.737 kPa; the selected fill-pump head is 7 m. A regulated gas boundary supplies 100 kPa cake filtration differential. That pressure is excluded from a vented liquid-fill pump head.

Candidate	Published duty / reason	Qualification
LKH60,190 mm,1425 rpm	Ordinary 90 m3/h transfers;source head 6.574 m,shaft 2.410 kW,NPSHr1.849 m	Water 20 C curve; duty-specific checks.
LKH60,200 mm,1425 rpm	L109 manifold duty;source head 8.208 m,shaft 2.901 kW	Selected 7 m required head; candidate, not final purchase.
Glycol candidates	Declared head/power/NPSH adjustment factors	Factors are engineering screens, not manufacturer-certified viscosity correction.

Control requirements follow the represented operation: temperature control at 21 C/4 C, metered per-module wash dose after the cold-water permissive, gas differential-pressure control during filtration, and level/flow/temperature/concentration checks for transfers and CIP. The process cannot be explained solely from the number of arrows or blocks. [V01,D21]

23 Utilities, collection and layout

System	Selected/calculated basis	Why it exists
Process glycol	488.183 kW coincident sensible peak;600 kW selected	E-501 plus salting-cooling overlap,20%allowance and rounding. Excludes complete FD and supplied-cold upstream feeds.
Extraction hot water	108.737 kW peak;140 kW selected	J-201 service separate from cleaning.
Primary cold traps	216.077 kW upper bound for 5 active industrial lots	5 x18 x2.40085; an upper bound, not synchronized sum of profile maxima or whole FD.
CIP	1 wt%NaOH,75 C;110 m3/h per active loop;4 circuits	Dairy-method analogy plus explicit project recipe and dispatch.
CIP heating	700 kW selected;638.611 kW2 h cold-start requirement	Selected 20 m3 stock heating and makeup. Return/equipment reheating and losses need separate data/allowance.
Layout	100 m x72 m preliminary allocation;4.5 m rack	Design coordinates for routes/access, not an actual site address or vendor-certified footprint.

Waste streams can share collection where their composition and discharge schedule are compatible. V021 groups alkaline/CIP liquid, acid/salt/wash filtrate and freeze-dryer condensate separately. Tank size follows cumulative inflow minus the selected export capacity, not the sum of all annual waste.

SPENTALK is scheduled separately from the six rinse drains. Wash header priming inventory is separated from the native per-round recipe. These details prevent a process boundary stream from disappearing during equipment translation. Combining liquid in a receiver does not calculate treatment, neutralization,pH,COD or BOD.

The preliminary allocation includes wet processing,wash banks,drying and utilities. Raw cold storage,final packaging/external dispatch and final equipment access need further allocation after package footprints. No real factory address has been selected. [D21,V05]

24 What has been verified and what remains assumed

Evidence	What it establishes	What it does not establish
Native readback,reopen and rerun	Correct saved settings,connections and repeatable stream results	Measured extraction,washing,dialysis or drying performance.
Mass/enthalpy closure	Consistency of represented components and energy accounting	True electrolyte enthalpy,pH or molecular quality.
Transport balances/reference checks	Conservative integration,dimensions and method implementation	Unique fitted collagen constitutive parameters.
Industrial conflict checks	The selected deterministic campaign fits the declared resource/time model	Real staffing,random downtime or guaranteed annual availability.
Published pump curve comparison	A candidate covers the declared hydraulic duty at source conditions	Final material,seal,motor,min-flow and glycol guarantee.

Native maximum apparent-component boundary residuals are approximately 4.49e-6 kg/batch for Dialysis and 5.18e-6 kg/batch for Washing. Saved-and-reopened stream-component differences were zero. These are accounting/implementation checks. The raw global PER_ERROR field was retained without treating its numeric value as a physical-performance certificate.

The assumptions that most affect the scientific conclusion

The release rate and cake permeability/hold-up determine the washing purity-recovery-time tradeoff. Frozen critical temperature, dry-layer resistance and shelf/product heat transfer determine the drying cycle. Empirical extraction yield and 21 C quality compatibility determine the upstream product interpretation. Membrane area/permeability are the missing physical link for the dialysis time.

How to present current maturity

The course design has a reproducible process and industrial engineering basis. Equipment quantities have scheduling and geometry explanations. Some performance inputs remain transferred or selected. State those inputs and sensitivities directly; do not describe every assumed margin as proven protection or add equipment simply to make the flowsheet look sophisticated.

25 A complete example to explain to a teammate

From literature to an industrial wash bank

Step	Trace
1. Process concept	L02 supports repeated washing as an alternative to dialysis; its tuna protocol motivates the route.
2. Project recipe	Select 8 rounds,2 L/kg current wet cake,15 min contact. Do not label these the paper’s exact optimum.
3. Retention and transport	h=.85 from a patent-informed choice; f_stagnant=.10; diffusion fromL05 plus length/hindrance assumptions.
4. Hydraulic property	L04 proxy k=(2.82-1.64)e-16 m2; derive alpha from assumed porosity/density.
5. Protein loss	Select f30=.001; calculate first-order release during contact and filtration.
6. Native result	Coupled model gives 1064.211 kg water and 9.708762 kg collagen after washing; compile component ratios into CAKEWASH SEP.
7. Industrial physical lot	Multiply inventory and active area 18 x:19.1558 m3 wash water and 216 m2 filter area/lot.
8. Module arrangement	14 modules/bank at 15.4286 m2 each preserve loading;2 banks support 6.805 h launches.
9. Utilities/control	E-501 prepares cold water;L109 fills through a balanced/metered manifold;gas controls cake differential.
10. Qualification	Actual thin-cake handling and fish-cake transport remain performance requirements, not measurements created by scale-up.

The same explanation pattern applies to every unit: physical objective, source data, selected assumptions, governing equations, computed requirement and concrete applicability limit. This is more informative than saying that the design follows a paper or that Aspen converged.

26 Native separation fractions

Each number below is the fraction of that component in the combined block inlet sent to the named retained/product outlet. It is not the component mass fraction of that outlet. The other outlet receives the complement. Values here are rounded to six significant figures; full precision and exact paths are in the parameter workbook. An absent component may have a zero fraction without implying a demonstrated removal performance.

Block	WATER	HAC	COLLAGEN	OTHERDRY	BASEEQ	NACLEQ
DRAIN	0.230779	0	0.948931	0.79931	0.230779	0
SKWASH6	0.0429724	0	0.983855	0.924679	9.13869e-05	0
CLARIFY	0.961437	0.961437	0.432154	0.182312	0.961437	0
SALTOUT	0.0441492	0.0441492	0.990441	0.139734	0.0441492	0.0441492
Washing CAKEWASH	0.0501246	0.00484924	0.980246	0.614233	0.000545174	0.000545174
Washing FREEZDRY	0.00932495	0.0100004	1	1	1	1
Dialysis DIALYSE	0.0466626	0.00194846	0.99	0.5	0.000718798	0.000718798
Dialysis FREEZDRY	0.000517227	0.01	1	1	1	1

Shared upstream values are shown once; the two route-specific drying water fractions differ because their feeds contain very different water inventories. FREEZDRY retains all NC equivalents. Acid fractions differ slightly from 0.01 because the apparent acid coordinate contains acetate-bound equivalent. [A01,D01]

27 Washing native block index

Native block	Type	Function
PRETREAT	MIXER	Alkaline contact/pretreatment
DRAIN	SEP	Drain spent alkaline liquor
SKWASH6	SEP	Six successive skin rinses
ACIDMIX	MIXER	Prepare 0.9 M extraction acid
ACIDCOOL	HEATER	Hold extraction acid at 21 C
EXTMIX	MIXER	Charge washed skin and extraction acid
EXTRACT	HEATER	Set extraction mixture to 21 C
CLARIFY	SEP	Recover collected acid extract
SALTMIX	MIXER	Add salt to collected extract
CHILL	HEATER	Cool salted mixture to 4 C
SALTOUT	SEP	Initial collagen precipitate separation
WATCOOL	HEATER	Cool washing water to 4 C
CAKEWASH	SEP	Eight reslurry/filtration cycles
FREEZDRY	SEP	Dry product to imposed endpoint

28 Dialysis native block index

Native block	Type	Function
PRETREAT	MIXER	Alkaline contact/pretreatment
DRAIN	SEP	Drain spent alkaline liquor
SKWASH6	SEP	Six successive skin rinses
ACIDMIX	MIXER	Prepare 0.9 M extraction acid
ACIDCOOL	HEATER	Hold extraction acid at 21 C
EXTMIX	MIXER	Charge washed skin and extraction acid
EXTRACT	HEATER	Set extraction mixture to 21 C
CLARIFY	SEP	Recover collected acid extract
SALTMIX	MIXER	Add salt to collected extract
CHILL	HEATER	Cool salted mixture to 4 C
SALTOUT	SEP	Initial collagen precipitate separation
REDMIX	MIXER	Redissolve salt cake
REDCHIL	HEATER	Cool redissolved collagen to 4 C
BATHMIX	MIXER	Prepare dialysis bath inventory
BATHCOOL	HEATER	Cool dialysis bath supplies to 4 C
DIALYSE	SEP	Reference finite-bath desalting
FREEZDRY	SEP	Dry product to imposed endpoint

For a15-20 min explanation, use:1)basis/routes,2)components and property limits,3)extractor and separator examples,4)washing/drying mechanisms,5)industrial scale and scheduling,6)source trace and assumptions. Keep the register open for detailed questions. Present the native batch and industrial lot numbers with their units every time.

29 References and exactly where they were used

L01 Scheja, Sigurdarson, Olafsson and Einarsson (2024). Maximizing Collagen Yield from Underutilized Lumpfish (Cyclopterus lumpus) Skins by Optimizing Pre-Cleaning and Extraction Methods. Marine Drugs 22, 525.

Location: Table 2, PDF p6; Sections 3.3-3.5, PDF p9.
Use: Raw composition, net pretreatment retention, extraction recipe, empirical collection ratios, nominal 2.6 M salt-out.
Limit: The selected composition branch and the optimized warm recipe are combined as a design scenario. This is not a reproduction of the complete pilot sequence.

https://doi.org/10.3390/md22120525

L02 Ampitiya, Gonapinuwala, Fernando and de Croos (2025). A simplified alternative to the conventional dialysis of the collagen extraction protocol. Food and Bioproducts Processing 152, 151-158.

Location: Sections 2.5.1-2.5.2 and 3.2; inherited downstream source ledger.
Use: Reference dialysis sequence and repeated reslurry washing as an alternative purification operation.
Limit: Yellowfin tuna, not lumpfish. Source protocol does not supply our 8-round transport parameters or equipment rating.

https://doi.org/10.1016/j.fbp.2025.05.008

L03 Vate, Strachowski, Undeland and Abdollahi (2023). Structural and functional properties of collagen isolated from lumpfish and starfish using isoelectric precipitation vs salting out. Food Chemistry: X 18, 100646.

Location: Sections 2.4.6, 3.7 and 3.8.
Use: Solution denaturation comparison (16.1 C) and relative salt-solubility context.
Limit: No frozen collapse temperature, absolute wash-release kinetics or plant salt-out isotherm.

https://doi.org/10.1016/j.fochx.2023.100646

L04 Hadjipanayi, E. (2010). Engineering physical structure in biomimetic collagen scaffolds: strategies for regulating cell behavior. PhD thesis, UCL.

Location: PDF pp 103 and 111-112.
Use: Permeability proxy: mean 2.82e-16 m2 and reported spread 1.64e-16 m2. Project stress selection uses their difference.
Limit: Compressed reconstituted collagen membrane, not precipitated fish cake; spread is not a statistical confidence interval.

https://discovery.ucl.ac.uk/id/eprint/624502/

L05 USGS PHREEQC 3, shipped phreeqc.dat diffusion data.

Location: Na+ and Cl- SOLUTION_SPECIES -dw records; frozen database hash in V006 literature_constraints.json.
Use: Dilute-ion diffusion at 25 C and temperature adjustment.
Limit: Hindrance, aggregate length, acid diffusion ratio and protein mobility are project assumptions.

https://www.usgs.gov/software/phreeqc-version-3

L06 Mahdi and Holdich (2013). Laboratory cake filtration testing using constant rate. Chemical Engineering Research and Design.

Location: Darcy and constant-pressure cake-filtration method; DOI record.
Use: Filtration equation and pressure-dependent resistance framework.
Limit: No calcium-carbonate resistance constant is used as a collagen measurement.

https://doi.org/10.1016/j.cherd.2012.11.012

L07 Murphy and Koop (2005). Review of the vapour pressures of ice and supercooled water for atmospheric applications. Quarterly Journal of the Royal Meteorological Society 131, 1539-1565.

Location: Equation 7.
Use: Pure-ice vapour pressure for the sublimation-front calculation.
Limit: Does not represent collagen-solution water activity.

https://doi.org/10.1256/qj.04.94

30 References and exactly where they were used (continued)

L08 Srisuma, Barbastathis and Braatz (2023). Analytical solutions for the modeling, optimization, and control of microwave-assisted freeze drying. Computers & Chemical Engineering 177, 108318.

Location: Conventional freeze-drying submodel and reference solution; V005 reference benchmark.
Use: Moving-front/thermal method comparison.
Limit: No microwave equipment is installed in this project. Reference material parameters are not collagen calibration.

https://web.mit.edu/braatzgroup/Srisuma_ComputChemEng_2023b.pdf

L09 Practical Advice on Scientific Design of Freeze-Drying Process: 2023 Update. Pharmaceutical Research.

Location: Primary drying, product critical temperature, heat/mass transfer and equipment capability.
Use: Method for relating product temperature and dry-layer resistance to a drying cycle.
Limit: Does not provide our formulation-specific critical temperature or resistance curve.

https://doi.org/10.1007/s11095-023-03607-9

L10 DE102005008416 B4. Process for purification of marine collagen.

Location: Description moisture-adjustment paragraph; Claims 1-2.
Use: Context for selecting 85% held mother liquor in wet cake.
Limit: Patent moisture range for another marine material; our 85% is a mother-liquor fraction, not a measured fish-cake water fraction.

https://patents.google.com/patent/DE102005008416B4/en

L11 de Queiroz et al. (2026), tilapia collagen ecodesign study.

Location: PDF p5 Section 2.2; PDF p9 Section 3.2.1 and supplementary material.
Use: 24 h salting-hold analogy; historical UF screening context.
Limit: Different fish/process. The 24 h hold is a transferred design selection, not lumpfish kinetic data.

https://doi.org/10.1007/s11367-026-02576-6

T01 ASHRAE Handbook - Refrigeration (2018), Chapter 19, Table 1; Choi-Okos generic food-component correlations.

Location: Protein Cp and density, temperature in C; coefficient conversion documented in fullplant/nc_representation.md.
Use: Generic protein sensible heat and density for COLLAGEN and OTHERDRY.
Limit: OTHERDRY composition is unresolved; generic protein is an analogue, not measured collagen properties.

https://handbook.ashrae.org/handbooks/R18/SI/R18_Ch19/r18_ch19_si.aspx

T02 Coulson and Richardson, Chemical Engineering, Vol. 2 B (2019), Chapter 10.

Location: pp 558-560, Eqs 10.2-10.12.
Use: Darcy/Ruth filtration and thickness-to-mass-specific resistance conversion.
Limit: Assumptions of constant pressure, viscosity and appropriate cake deposition must hold.

T03 Coulson and Richardson, Chemical Engineering (2018), Chapter 7.

Location: pp 343-349, Eq7.13 and Example 7.2.
Use: Impeller Reynolds number, power number and tip speed.
Limit: Chosen impeller geometry and Np are trial inputs; fibrous-slurry suspension is not established.

31 References and exactly where they were used (continued)

T04 Towler and Sinnott, Chemical Engineering Design, 3 rd edition (2022), Chapter 19.

Location: pp 826-827 Table 19.1; p848 Eq19.11; pp 934-939 Eqs 19.69-19.70, Examples 19.15-19.16.
Use: Thermal resistances, agitated-vessel jackets and convective heat transfer.
Limit: Published general ranges and correlations do not establish our actual U or rheology.

T05 CHEE3004 pressure-drop and heat-exchanger calculation workbooks.

Location: Week 1 Workshop Activity 2; methanol cooler corrected workbook; exact file records in V012/V014.
Use: Teaching workflow for Darcy-Weisbach, fittings, tube/shell geometry, U, area and pressure-drop checks.
Limit: Original files were preserved. Project calculations use corrected pass/diameter/unit logic.

T06 MIT OpenCourseWare, 1.060 Engineering Mechanics II, Spring 2006, Lecture 16.

Location: Darcy-Weisbach lecture.
Use: Single-phase pipe friction framework.
Limit: Water-like or declared Newtonian services only; not fibrous-slurry validation.

https://ocw.mit.edu/courses/1-060-engineering-mechanics-ii-spring-2006/3a800660e36bf54f7920afabc2cdd229_lecture16.pdf

P01 NIST Chemistry WebBook and NIST-JANAF solid heat-capacity data.

Location: NaCl Shomate 298-1073 K and NaOH Shomate 298-572 K; evaluated at 298.15 K.
Use: NC constant Cp analogues for NACLEQ and BASEEQ.
Limit: Solid-reference Cp frozen at 25 C is not an aqueous partial heat capacity or dissolution heat.

https://webbook.nist.gov/

P02 Official solid-density references compiled in fullplant/solute_properties.json.

Location: NaCl 2.16 g/cm 3 at 25 C; NaOH 2.13 g/cm 3 at 20 C.
Use: NC accounting density analogues.
Limit: Not bulk wet-cake or solution density.

P03 NIST Chemistry WebBook water Antoine/Stull correlations.

Location: 273-303 K coefficients and Stull 1947 255.9-373 K coefficients; units bar and K.
Use: Water vapour pressure in NPSHA, including 75 C CIP.
Limit: Acid/salt mixtures use an explicit upper-bound design input where selected.

https://webbook.nist.gov/cgi/cbook.cgi?ID=C7732185&Mask=4&Type=ANTOINE&Plot=on

V01 Alfa Laval LKH performance-curves product leaflet.

Location: PDF p19 LKH60 at 1425 rpm; other exact pages in V021 published-curve records.
Use: Published Q-H, Q-P and Q-NPSHr candidate checks.
Limit: Curves for water at 20 C; fluid corrections, stable minimum flow, seals and procurement configuration require confirmation.

https://assets.alfalaval.com/documents/p207424f4/alfa-laval-lkh-performance-curves-product-leaflet-en.pdf

32 References and exactly where they were used (continued)

V02 Dow DOWFROST Engineering and Operating Guide.

Location: Tables 10,14,18,22; V014 source excerpts.
Use: 30 vol% propylene glycol properties interpolated at-1 C.
Limit: Nominal mixture temperature/composition, not a measured installed-loop condition.

V03 Atlas Steels Product Reference Manual.

Location: Section 3 p6, ASTM A312 M/ASME B36.19 M schedule 40 S geometry.
Use: Pipe OD/wall and calculated bore.
Limit: Nominal geometry does not certify installed fittings or hydraulic layout.

https://www.atlassteels.com.au/documents/AtlasPRMcomplete.pdf

V04 Martin Christ production freeze-dryer catalog.

Location: Epsilon 2-220 D shelf/ice-capacity table; later industrial cabinet choices are project capacity requirements.
Use: Catalog feasibility screening only.
Limit: Maximum shelf/ice inventory is not a guarantee of our active area, peak vapour rate or cycle.

https://www.martinchrist.de/en/products/production-systems/bulk-and-vial-drying

V05 Tetra Pak Dairy Processing Handbook.

Location: Cleaning of dairy equipment: detergent concentration, mechanical effect and CIP programmes; long-life dairy products cleaning duration.
Use: CIP method analogy: concentration, temperature, velocity and selected 75 min allowance.
Limit: Dairy method does not validate collagen cleaning. Dose, bleed, rinse, stock and heater are project choices.

https://dairyprocessinghandbook.tetrapak.com/chapter/cleaning-dairy-equipment

V06 Merck Increase Product Yield in Your UF/DF Process, AN1026 EN00 v3.0 (2021); Pellicon 2 guide.

Location: Yield equation and Tables 1-4; feed-flow/TMP/flux-excursion sections.
Use: Historical UF/DF losses, holdup, adsorption and rinse framework.
Limit: No current UF block in Washing_0.4 or Dialysis_0.4. MWCO does not prove actual rejection.

https://www.merckmillipore.com/deepweb/assets/sigmaaldrich/product/documents/371/357/increase-product-yield-an1026en-mk.pdf

A01 Saved/reopened Aspen Plus V15 input exports and readbacks.

Location: V009 final reopened .inp files, streams_aspen_readback.csv and blocks_aspen_qc.csv.
Use: Exact active inputs, connectivity and calculated apparent stream results.
Limit: Agreement and convergence verify implementation, not independent process performance.

A02 Local Aspen Plus component databanks and property models.

Location: PURE41/AQUEOUS/SOLIDS/INORGANIC; NRTL-HOC with VLMX26; REFPROP for refrigeration.
Use: Molecular solvent properties and refrigerant-state calculations.
Limit: Binary parameter range includes a recorded 20 C lower bound;4 C use extrapolates. Installed parameters are not a new experimental fit.

33 References and exactly where they were used (continued)

D01 Project design assumptions and mass-balance compilation.

Location: Upstream/downstream ledgers, V007 physics, V009 exports.
Use: Explicit operating choices, empirical split assumptions and algebraic conversion to Aspen.
Limit: No external measurement is claimed for an assumed value.

D21 Revision 021 industrial design choices and dispatch.

Location: Design_Protocol.json, basis_product.json, integrated_design.json and scheduling/schedule_summary.json.
Use: 18 x physical lot,200 t/y target, equipment arrangement, campaign, layout, utilities and collection.
Limit: External physical design; normalized native process endpoints retained. Dialysis full industrial dispatch remains open.